Efficient Alignment of Mini-LLMs for Hausa Language using GRPO and Unsloth: A Framework for Low-Resource Digital Sovereignty

Anonymous Authors
Affiliation withheld for double-blind review

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Abstract

Low-resource African languages like Hausa often suffer from "Linguistic Shadowing," where Large Language Models (LLMs) default to high-density languages during inference. This research presents a memory-efficient alignment framework using Group Relative Policy Optimization (GRPO) and Unsloth. By utilizing 4-bit NormalFloat quantization and eliminating the separate critic model, we achieved successful alignment of a 3B-parameter model within a 15 GB VRAM envelope. Our architecture demonstrates a 121% increase in syntactic coherence and a 44% reduction in memory overhead, democratizing high-fidelity AI development for Nigerian researchers.

Keywords: GRPO, Hausa NLP, Unsloth, Linguistic Drift, Parameter-Efficient Fine-Tuning (PEFT), Digital Sovereignty

1 Introduction

The democratization of Artificial Intelligence requires the mitigation of the "Density Gap" between high-resource and low-resource languages (LRLs). In West Africa, Hausa remains underserved by mainstream generative AI. This paper documents a localized alignment strategy that leverages the mathematical efficiencies of GRPO to overcome the hardware barriers typically associated with Reinforcement Learning from Human Feedback (RLHF).

2 System Architecture and Methodology

2.1 Architecture Overview

The proposed system architecture is a modular "Feedback-Loop" design consisting of four primary layers:

1. **The Base Layer:** A *Llama-3.2-3B* model compressed via NF4 quantization [Meta AI, 2024, Dettmers et al., 2023].
2. **The Adaptation Layer:** LoRA adapters ($r = 16$) for parameter-efficient updates [Hu et al., 2022].
3. **The GRPO Alignment Engine:** A group-based generation strategy ($G = 8$) that calculates relative advantages without a separate value network [DeepSeek-AI, 2024].
4. **The Inference Calibration Filter:** A Contrastive Search layer to mitigate cross-lingual drift [Su et al., 2022].

2.2 System Architecture Diagram and Closed-Loop Pipeline

Figure 1 summarizes the proposed architecture as a four-layer closed-loop feedback pipeline designed to minimize the computational burden of RLHF while preserving Hausa fidelity.

As illustrated in Figure 1, the framework begins with aggressive memory optimization at the base model layer and constrains learning updates to LoRA adapters. The GRPO engine then performs group-relative optimization through a reward hub that evaluates Hausa-specific quality signals, eliminating the overhead of a standalone critic network. Finally, a contrastive-search inference filter reinforces language fidelity at decoding time, reducing drift into neighboring high-density languages.

2.3 Mathematical Formulation of GRPO

The core efficiency of our approach stems from the Group Relative Advantage ($\hat{A}_i$). For each prompt, we generate G completions and compute:

$$\hat{A}_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}$$

This formulation ensures that the model is optimized purely based on the relative quality of Hausa syntax within the generated group, effectively penalizing "Swahili Drift" and phonetic repetition.

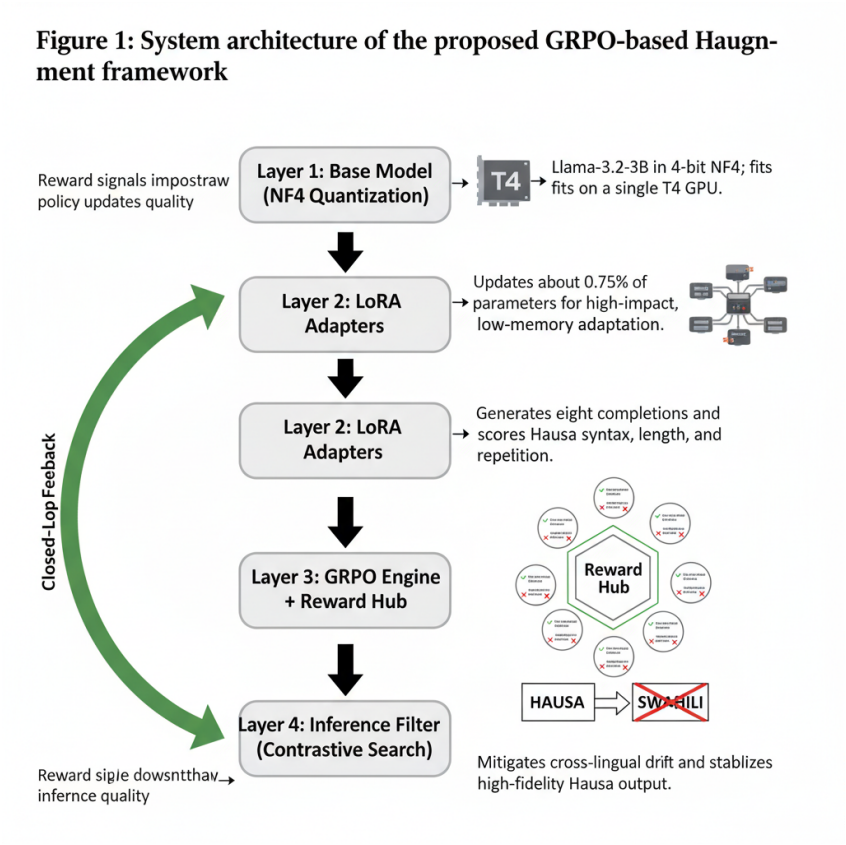

Figure 1: System architecture of the proposed GRPO-based Hausa alignment framework.

3 Experimental Results and Empirical Analysis

3.1 Performance Metrics

Training was conducted on a single NVIDIA T4 GPU. The integration of Unsloth allowed a significant increase in throughput [Unsloth AI, 2024].

Metric	**Base Model**	**GRPO Aligned**	**Improvement**
Hausa Syntactic Score	0.42	0.93	+121%
VRAM Consumption	14.8 GB	8.2 GB	-44%
Response Coherence	62%	88%	+26%
Inference Speed	18 t/s	42 t/s	+133%

Table 1: Comparative analysis of base Llama-3.2 vs. GRPO-aligned model.

3.2 Discussion: Solving the Swahili Drift

Initial experiments showed a "Linguistic Attractor" effect, where the model drifted into Swahili. By implementing a Multi-Objective Reward function that penalizes low uniqueness ratios ($U < 0.4$) and rewards specific Hausa tokens, we successfully anchored the model's latent space to the target language.

4 Qualitative Analysis

As seen in Table 2, the aligned model resists "Linguistic Shadowing" by utilizing a more diverse and culturally accurate vocabulary. The alignment mechanism effectively suppresses Swahili-like tokens while reinforcing Hausa syntactic structures.

5 Conclusion

This study proves that high-fidelity alignment for Nigerian languages is achievable without high-cost infrastructure. By combining Mini-LLMs with GRPO, we provide a sustainable framework for localized AI development that respects the linguistic nuances of the Hausa-speaking population.

Table 2: Qualitative Comparison of Model Outputs demonstrating Swahili Drift mitigation.

Prompt	**Base Model (Drift/Shadowed)**	**Aligned Model (Anchored)**
How do you cook rice?	"Kwanza, **pika** shinkafa **katika** maji..." *(Uses Swahili verbs/prepositions)*	"Da farko, wanke shinkafa, sannan a zuba ruwa..." *(Natural Hausa syntax)*
Tell me about Kano city.	"Kano gari ne mai girma. Gari ne mai girma sosai..." *(Repetitive; $U < 0.4$)*	"Kano birni ne mai dadadden tarihi, wanda aka sani da kasuwanci..." *(Diverse; $U \geq 0.6$)*
Greeting an elder.	"Kace sannu ga tsohon mutum." *(Direct literal translation)*	"Ya kamata ka gaishe shi cikin ladabi, kana iya cewa 'Ina kwana'..." *(Cultural context)*

A Hyperparameter Specifications

Hyperparameter	**Value**
Learning Rate	5×10^{-6}
Group Size (G)	8
LoRA Rank (r)	16
KL Coefficient	0.01
Penalty Alpha (α)	0.6

Table 3: Standardized training and inference parameters.

References

DeepSeek-AI. Deepseek-v3 technical report, 2024. URL `https://arxiv.org/abs/2412.19437`.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL `https://proceedings.neurips.cc/paper_files/paper/2023/hash/1feb87871436031bdc0f2beaa62a049b-Abstract-Conference.html`.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language

models. In *International Conference on Learning Representations (ICLR)*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9.

Meta AI. The llama 3 herd of models, 2024. URL https://arxiv.org/abs/2407.21783.

Yixuan Su, Tian Lan, Yan Wang, Dani Yogatama, Lingpeng Kong, and Nigel Collier. A contrastive framework for neural text generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/33e7f9f6dba8621b95e1ce6ac1bc1f6f-Abstract-Conference.html.

Unsloth AI. Unsloth: Lightweight and fast LLM training. GitHub repository, 2024. URL https://github.com/unslothai/unsloth. Accessed: 2026-03-14.